

What Is Machine Learning?

Learning from experience, verification, and the real world

Machine learning uses experience to improve decisions. Improvement depends on evidence about performance, failure, and transfer to the intended environment. This lecture connects building, verification, and real-world use.

Learning goals. By the end of this lecture, you should be able to

1. define machine learning and distinguish a learned model from a fixed program;
2. explain how verification connects building to evidence; and
3. explain why real-world use requires evidence from the target environment.

1 Building shifts the work

Traditional engineering often begins by learning the technology: *How does it work?* AI tools can produce a prototype earlier. Engineering effort then shifts to three review questions:

Why should it work? When will it fail? What evidence supports it?

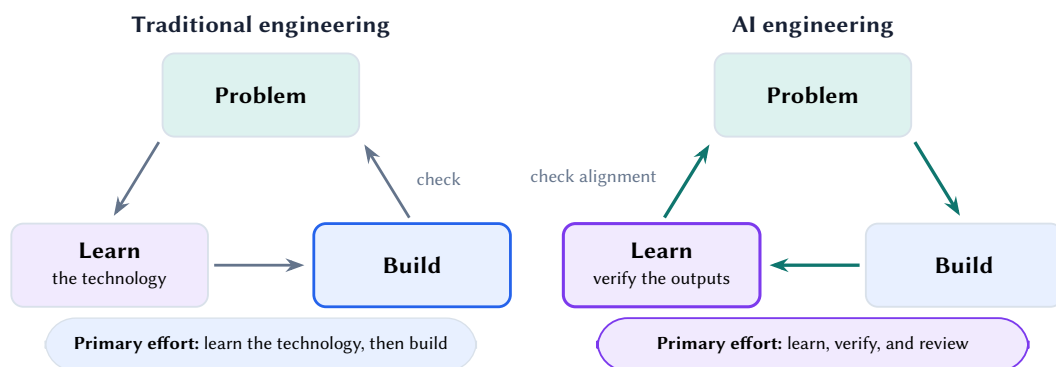


Figure 1. Traditional engineering learns before building. AI engineering builds earlier and shifts effort toward learning from outputs, verification, review, and alignment.

The AI cycle studies the outputs and checks their alignment with the original problem. Review and verification guide the next build.

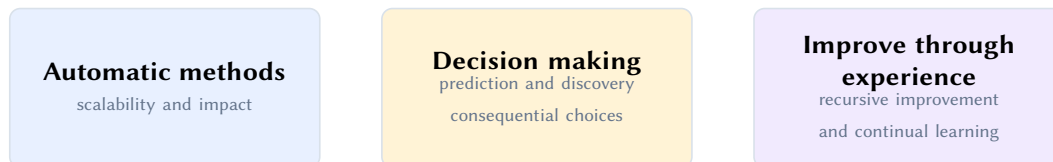
Start with the intended use. Define the task, operating environment, and cost of failure before choosing a model. These choices determine

the evidence required. Unit tests measure software behavior under specified cases. Medical and control systems also require evidence from their operating conditions.

People set the goal, identify important failures, collect evidence, and decide whether the evidence supports the intended use.

2 Machine learning uses experience

A practical definition has three parts:



Automation gives machine learning scale and impact. Predictions and discoveries can shape consequential choices. Improvement requires experience and a feedback signal aligned with the goal; repeated feedback can support continual learning and recursive improvement.

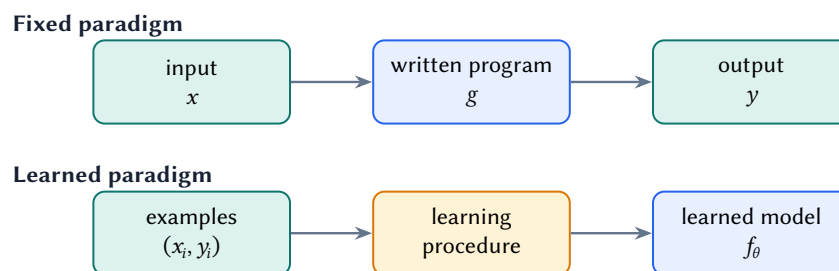


Figure 2. A fixed program applies written rules to data. A learned system uses examples and a learning procedure to construct the model.

Figure 2 uses labeled examples. Other systems learn from rewards, comparisons, demonstrations, or unlabeled data. For each setting, identify the experience, the part of the system that changes, and the measure of improvement.

Machine-learning history is cumulative. Early neural and neighbor methods remain useful. Later advances expanded the scale of learning and changed how people interact with learned systems.

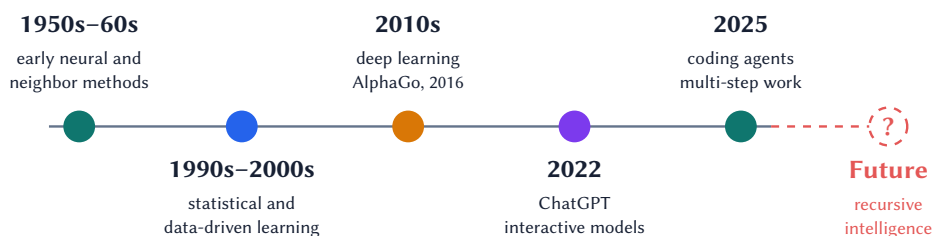


Figure 3. Selected milestones and one open question about the future of machine learning.

ChatGPT made foundation models widely interactive. Coding agents extend them into multi-step work. Recursive intelligence remains an open question. These systems require clear objectives, review, and evidence.

3 Learning needs a verification signal

Generation produces candidates. A verification signal measures behavior, judges improvement, and guides the next attempt.

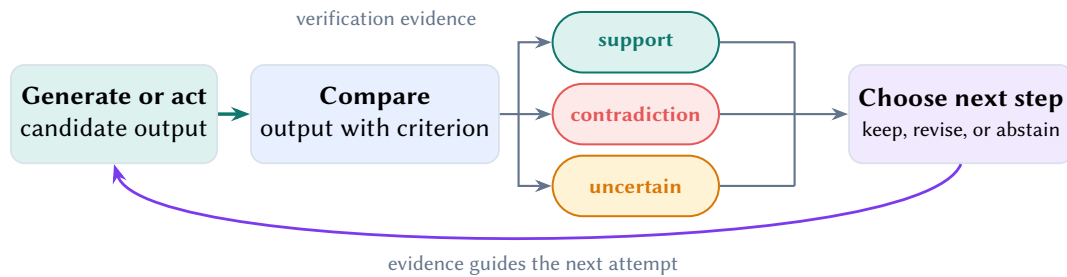


Figure 4. Verification maps evidence to an outcome, then uses that outcome to choose the next action.

Software tests, checked calculations, and game scores provide direct digital feedback. Each check measures a specific criterion.

Setting	Candidate	Possible signal	What remains unresolved
Software	Program or patch	Tests, static checks, runtime behavior	Whether the tests express the full requirement
Mathematics	Answer or proof	Recalculation, proof checking, counterexamples	Whether the assumptions and question are correct
Physical system	Decision or policy	Sensors, constraints, outcomes over time	Noise, drift, delayed effects, and rare harm

Table 1. The useful verification signal depends on the task. Every signal leaves some uncertainty outside what it measures.

Match the signal to the intended use

Tests cover tested behavior. Scores encode the chosen objective. Sensors cover what they observe. Choose criteria for the intended use and major failure modes.

Trust grows from repeated evidence for a specific claim under stated conditions. Missing, ambiguous, or out-of-environment evidence weakens the claim.

4 Real-world use requires real-world evidence

Digital environments offer repeatable trials and fast feedback. Real-world use adds hidden state, measurement error, changing conditions, people, and irreversible consequences.

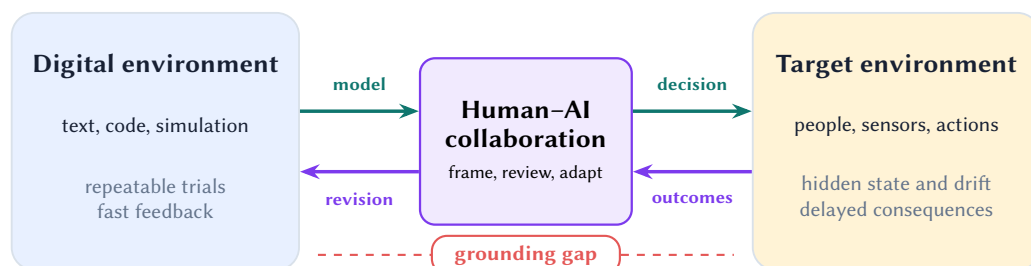


Figure 5. Human-AI collaboration helps bridge the grounding gap. People frame the use, review evidence from the target environment, and guide revision.

Digital training captures patterns in text, code, simulations, or games. Deployment also depends on physical causes, human behavior, and real consequences. Simulation narrows the gap. Target-environment evidence tests the remaining mismatch.

Human–AI collaboration keeps the transfer claim explicit. AI supports rapid generation and simulation. People define the task, interpret target-environment evidence, and decide what requires revision.

Diagnose failures through the evidence. Inspect the problem formulation, training experience, verification signal, and deployment environment before tuning the learning algorithm.

Lecture summary. Machine learning uses experience to improve decisions. Relevant evidence guides each iteration and tests whether performance transfers to the intended environment.

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